

# SkillCraft Technology Internship

## Task 01: Writing Better Prompts

Prepared by: Shaik Suheal

### Introduction

Prompt Engineering is the process of creating effective instructions for AI systems. The quality of a prompt directly impacts the quality of the AI-generated response. This report demonstrates how vague prompts can be improved to generate more detailed and useful outputs.

### Example 1

**Vague Prompt:** Explain Artificial Intelligence.

**AI Output:** Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that usually require human intelligence.

**Improved Prompt:** Explain Artificial Intelligence in 8 bullet points for MCA students. Include applications, advantages, disadvantages, and examples.

**Improved AI Output:**

1. AI enables machines to mimic human intelligence.
2. AI systems learn from data and improve performance.
3. Used in healthcare, education, finance, and business.
4. Powers virtual assistants such as ChatGPT and Siri.
5. Improves automation and productivity.
6. Advantages include speed, accuracy, and efficiency.
7. Disadvantages include ethical concerns and job displacement.
8. Examples include recommendation systems and self-driving cars.

**Analysis:** The improved prompt provides context, structure, and expectations, resulting in a more comprehensive response.

### Example 2

**Vague Prompt:** Tell me about Python.

**AI Output:** Python is a popular programming language used for software development and data analysis.

**Improved Prompt:** Explain Python programming for beginners in 10 bullet points with practical examples and career opportunities.

**Improved AI Output:**

1. Python is easy to learn and read.
2. Used in web development.
3. Popular in data science and AI.
4. Supports automation tasks.
5. Has a large developer community.
6. Provides extensive libraries.
7. Used for machine learning projects.
8. Example: Building web apps with Django.
9. Example: Data analysis using Pandas.
10. Career roles include Python Developer and Data Analyst.

**Analysis:** The improved prompt provides context, structure, and expectations, resulting in a more comprehensive response.

### Example 3

**Vague Prompt:** What is Cloud Computing?

**AI Output:** Cloud computing delivers computing services over the internet.

**Improved Prompt:** Explain Cloud Computing with service models, deployment models, benefits, and examples.

**Improved AI Output:**

1. Cloud computing provides on-demand computing resources.
2. Service models include IaaS, PaaS, and SaaS.
3. Deployment models include public, private, and hybrid cloud.
4. Reduces infrastructure costs.
5. Provides scalability and flexibility.
6. Improves accessibility.
7. Supports remote work.
8. Examples include AWS, Azure, and Google Cloud.

**Analysis:** The improved prompt provides context, structure, and expectations, resulting in a more comprehensive response.

## **Key Learnings**

- Specific prompts generate better outputs.
- Defining the audience improves relevance.
- Structured instructions produce organized responses.
- Prompt engineering is essential for effective AI interaction.

## **Conclusion**

This task demonstrated that carefully crafted prompts significantly improve AI response quality. Adding context, audience, format, and objectives helps obtain accurate and meaningful results.